

AN ALMOST-SURE EXPANSION FOR REGRESSION M-ESTIMATORS

J.L. Fresen: Department of Statistics, University of the Western Cape,
Private Bag X17, Bellville, 7535.

and

F. Lombard: Department of Statistics, Rand Afrikaans University, P.O.
Box 524, Johannesburg, 2000.

Key words: Almost-sure expansion; M -estimators; regression.

Summary: In this paper we derive an almost-sure expansion for regression M -estimators, generalizing results of Carroll (1978). The results, which apply to independent (and some types of dependent) data sequences, have been found useful in the study of robust regression residual processes.

1. INTRODUCTION

Carroll (1978) derived an almost-sure expansion for M -estimators for functionals of the form $\theta = \theta(F)$, where F is a probability distribution function, in the one sample setting. The object of this paper is to obtain an almost-sure expansion for M -estimators in the regression setting. Specifically, consider the linear regression model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \sigma\boldsymbol{\epsilon}$ which, in component form, is written as $Y_i = \mathbf{z}_i^t \boldsymbol{\beta} + \sigma\epsilon_i$, ($i=1, \dots, n$). Thus, Y_i denotes the i -th response (dependent variable), $\mathbf{z}_i^t = (1, x_{i1}, x_{i2}, \dots, x_{ip})$ a vector of covariates (the i -th row of the design matrix $\mathbf{X}$), $\boldsymbol{\beta}^t = (\beta_0, \beta_1, \dots, \beta_p)$, a $(p+1)$ vector of unknown regression parameters, ϵ_i the i -th error, and σ a scale parameter. The errors are assumed to be independent and identically distributed (iid) according to some unknown *symmetric* distribution F with a mean of 0 and variance 1.

STMA 1992 SUBJECT CLASSIFICATION: 04:110(04:090)

To estimate β , we consider solutions $\hat{\beta} \equiv \hat{\beta}_n$ of the equations

$$(1) \quad \sum_{i=1}^n \psi \left[\frac{Y_i - \mathbf{x}_i^t \hat{\beta}}{s_n} \right] \mathbf{x}_i = 0$$

where ψ is a bounded function whose properties will shortly be specified and s_n is a translation invariant and scale equivariant consistent estimate of σ . In Theorem 1 conditions are given under which the error of estimation $(\hat{\beta}_n - \beta)$ can be expressed almost surely as

$$\hat{\beta}_n - \beta = \sigma \nu^{-1} \cdot \frac{1}{n} \sum_{i=1}^n \psi(\epsilon_i) \mathbf{B}^{-1} \mathbf{x}_i + O(n^{-1} \log_2 n)$$

where ν is a constant and $\mathbf{B}$ a matrix, both of which will be defined later.

The motivation for this result is the detection of change-points in a regression setting. Consider, for instance, the model

$$Y_i = \alpha + \gamma(i-m)I(i \leq m) + \epsilon_i$$

with $1 \leq m < n$ and $I(\cdot)$ denoting the indicator function. Here m is unknown and interest centres on (i) testing $H_0: \gamma = 0$ and (ii) estimating m if $\gamma \neq 0$. Here we shall restrict attention to problem (i). One approach is to fit a model $Y_i = \beta_0 + \beta_1 i + \epsilon_i$ and to examine the resulting residual process

$$\sum_{i=1}^k (Y_i - \hat{\beta}_0 - \hat{\beta}_1 i).$$

Fresen *et al.* (1992) use the above representation for $\hat{\beta}_0$ and $\hat{\beta}_1$ to derive functional limit theorems (invariance principles) involving this residual process based on regression M -estimators. Another (sequential) approach is to estimate β_1 on the data $Y_1, \dots, Y_k$ for $k = 1, 2, \dots$, and denoting the estimates by $\hat{\beta}_{1k}$, to reject H_0 as soon as $|\hat{\beta}_{1k}|$ becomes "too large". For this one requires the distribution of $\max_{1 \leq k \leq n} |\hat{\beta}_{1k}|$. The almost sure

representation above can be used to show that, as $n \rightarrow \infty$,

$$C \cdot n^{\frac{3}{2}} \max_{1 \leq k \leq n} |\hat{\beta}_{lk}| \rightarrow \sup_{0 \leq t \leq 1} \left| \int_0^t x W(dx) \right|$$

in distribution, where W is a standard Brownian motion and C is a constant depending on the underlying distribution of ϵ and which is to be estimated from the data.

2. MAIN RESULTS

The estimating equation (1) can be written as

$$(2) \quad \sum_{i=1}^n \psi\{u_n(\epsilon_i - x_i^t \hat{\theta})\} x_i = 0$$

where $u_n = \frac{\sigma}{s_n}$ and $\hat{\theta} = (\hat{\beta} - \beta)/\sigma$. We shall assume throughout that ψ is skew symmetric. Then, using symmetry of F ,

$$(3) \quad E\psi\left[\frac{Y_i - x_i^t \beta}{\sigma}\right] = E[\psi(\epsilon_i)] = 0.$$

Let ψ' and ψ'' denote the first and second derivatives of ψ . Consider the following assumptions:

(A1) ψ is continuous and bounded and there are intervals $(-\infty, a_1), \dots, (a_k, \infty)$ on which ψ has two continuous bounded derivatives.

(A2) F satisfies a Lipschitz condition of order one in neighbourhoods of $\{a_1, \dots, a_k\}$ and $E_F[\psi'(\epsilon_1)] \neq 0$.

(A3) $\sup x^2 \psi''(x) < \infty$ and $E_F[\epsilon_1^2 \{\psi'(\epsilon_1)\}^2] < \infty$. The supremum is taken over the intervals mentioned in (A1).

(A4) The sequence of statistics $u_n = \frac{\sigma}{s_n}$ satisfies

$$\limsup_n n^{\frac{1}{2}} (\log_2 n)^{-\frac{1}{2}} |u_n - 1| < C_u \text{ a.s.,}$$

for some constant C_u . In the sequel we take $C_u = 1$ without loss of generality.

(A5) $\mathbf{x}_i = (f_0(\frac{i}{n}), \dots, f_p(\frac{i}{n}))^t$

where the functions f are each of polynomial form.

Remark: Since the functions in (A5) are bounded, we may assume without loss of generality that $\sup\{f_k(x): 0 \leq x \leq 1\} = (p+1)^{-\frac{1}{2}}$. For any $(p+1) \times 1$ column vector $\mathbf{u} = (u_0, \dots, u_p)^t$, write

$$\|\mathbf{u}\| = \left[\sum_{i=0}^p u_i^2 \right]^{\frac{1}{2}},$$

and observe that

$$\begin{aligned} (5) \quad \left| \mathbf{x}_i^t \cdot \mathbf{u} \right| &= \left| \sum_{k=0}^p f_k\left(\frac{i}{n}\right) u_k \right| \\ &\leq \left[\sum_{k=0}^p f_k^2\left(\frac{i}{n}\right) \right]^{\frac{1}{2}} \cdot \|\mathbf{u}\| \\ &\leq \|\mathbf{u}\| \end{aligned}$$

by Schwarz's inequality. ■

The proofs of Lemma 1 and Lemma 2 below follow a path laid out by Huber (1981, section 7.4) and Carroll (1978).

Lemma 1: Let conditions (A1) - (A5) hold and define $\gamma_n = (n^{-1} \log_2 n)^{\frac{1}{2}}$. Then there exist positive numbers C_1 and δ such that for any sequence δ_1 ,

$\delta_2, \dots$ in $[0, \delta]$ satisfying $\gamma_n \max_j |a_j| < (1 - \gamma_n) \delta_n$ for sufficiently large n , the following holds, viz. for almost all sequences ω there exists $n(\omega)$ such that

$$n \geq n(\omega) \text{ and } \|u\| \leq \delta_n$$

imply

$$(6) \quad \left\| \frac{1}{n} \sum_{i=1}^n \psi(u_n(\epsilon_i - x_i^t u)) x_i - \frac{1}{n} \sum_{i=1}^n \psi(\epsilon_i) x_i + B u \cdot \nu \right\| \\ \leq C_1 \left\{ (\|u\| + \gamma_n)^2 + (\|u\| + \gamma_n) \left(\delta_n \vee \frac{\log n}{n} \right) \right\}$$

and

$$(7) \quad \left\| \frac{1}{n} \sum_{i=1}^n \psi(u_n(\epsilon_i - x_i^t u)) x_i + B u \cdot \nu \right\| \leq C_1 (\|u\| \delta_n + \gamma_n)$$

where $\nu \equiv E_F \psi'(\epsilon_1)$ and $B \equiv \lim_{n \rightarrow \infty} n^{-1} X^t X$ is the matrix with (k, l) -th element

$$\int_0^1 f_k(x) f_l(x) dx.$$

Proof: The proof follows that of Lemma 2 of Carroll (1978). For instance, with his definition of $n(\omega)$, δ , B_n , δ_n , we have for $\epsilon \in B_n$, $\epsilon \neq a_j$ ($j = 1, \dots, k$), and for $r = 0, \dots, p$, that

$$\begin{aligned} & |\psi(u_n(\epsilon - x_{ir}^t u)) x_{ir} - \psi(\epsilon) x_{ir} - (u_n - 1) \epsilon \psi'(\epsilon) x_{ir} + u_n \psi'(\epsilon) x_{ir}^t u \cdot x_{ir}| \\ & \leq C_1 |\{u_n(\epsilon - x_{ir}^t u) - \epsilon\} x_{ir}| \\ & = C_1 |(u_n - 1) \epsilon - u_n x_{ir}^t u| |x_{ir}| \\ & \leq C_1 \{|u_n - 1| + \|u\|\}, \\ & \leq C_1 \{\gamma_n + \|u\|\}, \end{aligned}$$

using (5) and the boundedness of x_{ir} , and where C_1 denotes a generic constant. Also, the law of the iterated logarithm holds for the sequences

$\{\psi(\epsilon_i) x_{ir}\}$, $\{\epsilon_i \psi'(\epsilon_i) x_{ir}\}$ and $\{\psi'(\epsilon_i) x_{ir}\}$. To see this, recall the assumption that $x_{ir} = f_r(\frac{i}{n})$ is a polynomial, say $x_{ir} = (\frac{i}{n})^q$ for some positive integer q . Then, for instance, $\psi(\epsilon_i) x_{ir} = n^{-q} i^q \psi(\epsilon_i)$ and since

$$\begin{aligned} & n \cdot \text{var}\{n^q \psi(\epsilon_i)\} / \sum_1^n \text{var}\{i^q \psi(\epsilon_i)\} \\ & \sim n \cdot n^{2q} / \{n^{2q+1} (2q+1)^{-1}\} \\ & = O(1), \end{aligned}$$

it follows from corollary 6 on page 352 of Chow and Teicher (1978) that

$$\sum_{i=1}^n i^q \psi(\epsilon_i) = O\{n^{q+\frac{1}{2}} (\log_2 n)^{\frac{1}{2}}\} \quad \text{a.s.}$$

Hence,

$$(8) \quad \frac{1}{n} \sum_{i=1}^n x_{ir} \psi(\epsilon_i) = O\{(n^{-1} \log_2 n)^{\frac{1}{2}}\} = O(\gamma_n) \quad \text{a.s.}$$

Thus we obtain, following Carroll (1978), that

$$\begin{aligned} (9) \quad & \left\| \frac{1}{n} \sum_{i=1}^n \psi(u_n(\epsilon_i - x_i^t u)) x_i - \frac{1}{n} \sum_{i=1}^n \psi(\epsilon_i) x_i + \frac{1}{n} \sum_{i=1}^n x_i^t u \cdot x_i \nu \right\| \\ & \leq C_1 \left\{ (\|u\| + \gamma_n)^2 + (\|u\| + \gamma_n) (\delta_n \vee \frac{\log n}{n}) \right\} \end{aligned}$$

for $n \geq n(\omega)$ and $\|u\| \leq \delta_n$. Next, using Lemma P5.1 of Brillinger (1975, page 415), we find

$$\begin{aligned} & \frac{1}{n} \sum_{i=1}^n x_i^t u \cdot x_{ir} = \frac{1}{n} \sum_{i=1}^n \sum_{k=0}^p f_k(\frac{i}{n}) u_k f_r(\frac{i}{n}) \\ & = \sum_{k=0}^p u_k \cdot \frac{1}{n} \sum_{i=1}^n f_k(\frac{i}{n}) f_r(\frac{i}{n}) \\ & = \sum_{k=0}^p u_k \cdot \left[\int_0^1 f_k(\xi) f_r(\xi) d\xi + O(\frac{1}{n}) \right] \end{aligned}$$

$$\begin{aligned}
&= \sum_{k=0}^p u_k \cdot \int_0^1 f_k(\xi) f_r(\xi) d\xi + O\left(\frac{1}{n}\right) \\
&= (r\text{-th element of the column vector } \mathbf{Bu}) + O\left(\frac{1}{n}\right),
\end{aligned}$$

and since $n^{-1} < n^{-1} \log_2 n = \gamma_n^2$ for sufficiently large n , (6) follows from (9). Again, (7) follows from (6) by virtue of (8). ■

Lemma 2: Under the conditions of Lemma 1 there exists for every $n \geq n(\omega)$ a solution $\hat{\theta}_n$ of (2) which satisfies $\|\hat{\theta}_n\| < C_* \gamma_n$ where C_* is a constant not depending on n or ω .

Proof: First make the transformation $\mathbf{v} = \mathbf{B}\mathbf{u}$, that is $\mathbf{u} = \mathbf{B}^{-1}\mathbf{v}$. Then

$$\begin{aligned}
\|\mathbf{u}\|^2 &= \mathbf{u}^t \mathbf{u} \\
&= \nu^{-2} \mathbf{v}^t \mathbf{B}^{-2} \mathbf{v} \\
&= \nu^{-2} \sum_{i=0}^p \sum_{j=0}^p v_i v_j (\mathbf{B}^{-2})_{ij} \\
&= \nu^{-2} \sum_{i=0}^p \sum_{j=0}^p v_i v_j \sum_{k=0}^p b^{ik} b^{kj} \\
&\leq \nu^{-2} \sum_{i=0}^p \sum_{j=0}^p |v_i| |v_j| \sum_{k=0}^p |b^{ik}| |b^{kj}| \\
&\leq \nu^{-2} (p+1)^2 \|\mathbf{v}\|^2 \cdot (p+1) M^2
\end{aligned}$$

where $M = \max\{|b^{ik}| : 0 \leq i, k \leq p\}$. Thus, there is a constant C_2 such that $\|\mathbf{u}\| \leq C_2 \|\mathbf{v}\|$. Now choose $\|\mathbf{v}\| \leq 2C_1 \gamma_n$ and $\delta_n = \min(\frac{1}{2C_1 C_2}, \delta)$ for all n . Then $\|\mathbf{u}\| \leq 2C_1 C_2 \gamma_n (< \delta_n$ for sufficiently large n) and so, by (7) of lemma 1,

$$\begin{aligned}
& \left\| \frac{1}{n} \sum_{i=1}^n \psi \{ u_n(\epsilon_i - \mathbf{z}_i^t \mathbf{B}^{-1} \mathbf{v} \nu^{-1}) \} \mathbf{x}_i + \mathbf{v} \right\| \\
& \leq C_1 \{ 2C_1 C_2 \gamma_n \cdot \frac{1}{2C_1 C_2} + \gamma_n \} \\
& = 2C_1 \gamma_n.
\end{aligned}$$

Setting

$$(10) \quad g_n(\mathbf{v}) = \frac{1}{n} \sum_{i=1}^n \psi \{ u_n(\epsilon_i - \mathbf{z}_i^t \mathbf{B}^{-1} \mathbf{v} \nu^{-1}) \} \mathbf{x}_i,$$

we see that we have proved the following for every $n \geq n(\omega)$, viz.

$$(11) \quad \|\mathbf{v}\| \leq 2C_1 \gamma_n \text{ implies } \|g_n(\mathbf{v}) + \mathbf{v}\| \leq 2C_1 \gamma_n.$$

Now the map $h: \mathbb{R}^{p+1} \rightarrow \mathbb{R}^{p+1}$ defined by

$$h(\mathbf{v}) = g_n(\mathbf{v}) + \mathbf{v}$$

is continuous (because ψ is continuous) and (11) says that h maps the closed sphere $\{\|\mathbf{v}\| \leq 2C_1 \gamma_n\}$ into itself. Thus, by the Fixed Point Theorem (Dunford and Schwartz, 1963, page 453) there exists a point $\mathbf{v}_0 \in \{\|\mathbf{v}\| \leq 2C_1 \gamma_n\}$ such that $h(\mathbf{v}_0) = \mathbf{v}_0$, i.e. $g_n(\mathbf{v}_0) = 0$. Setting $\hat{\theta} = \mathbf{B}^{-1} \mathbf{v}_0 \nu^{-1}$ in (10) we see that (2) is satisfied. Furthermore, since $\hat{\theta} = \mathbf{B}^{-1} \mathbf{v}_0 \nu^{-1}$, the computation that started this proof shows that

$$\|\hat{\theta}\| \leq C_2 \|\mathbf{v}\| \leq 2C_1 C_2 \gamma_n = C_* \gamma_n. \quad \blacksquare$$

Theorem 1: Let $\{\hat{\theta}_n = (\hat{\beta}_n - \beta)/\sigma\}$ be any sequence of $\mathcal{O}(\gamma_n)$ a.s. solutions of (2), whose existence is guaranteed by Lemma 2. Then,

$$\hat{\beta}_n - \beta = \sigma \nu^{-1} \cdot \frac{1}{n} \sum_{i=1}^n \psi(\epsilon_i) \mathbf{B}^{-1} \mathbf{x}_i + \mathcal{O}(n^{-1} \log_2 n) \text{ a.s.}$$

Proof: Let $C^* = \max\{C_*, 2\max_j |a_j|\}$, where C_* is the constant in Lemma 2, and set $\delta_n = \min\{\delta, C^* \gamma_n\}$. Then for sufficiently large n

$$\delta_n = C^* \gamma_n \geq 2\max_j |a_j| \cdot \gamma_n > \max_j |a_j| \cdot \gamma_n (1 - \gamma_n)^{-1}$$

and

$$\|\hat{\theta}_n\| \leq C_* \gamma_n \leq C^* \gamma_n = \delta_n.$$

Thus we may use (6) with $\mathbf{u} = \hat{\theta}_n$. Since (2) holds, we get for $n \geq n(\omega)$

$$\begin{aligned} & \left\| B \hat{\theta}_n \nu - \frac{1}{n} \sum_{i=1}^n \psi(\epsilon_i) \mathbf{x}_i \right\| \\ & \leq C_1 \{(C^* \gamma_n + \gamma_n)^2 + (C^* \gamma_n + \gamma_n) C^* \gamma_n\} \\ & = O(n^{-1} \log_2 n) \text{ a.s. } \blacksquare \end{aligned}$$

Remarks:

(1) The preceding results have been proved under the assumption that f is a polynomial. This assumption can be weakened at the expense of a stronger assumption on the form of the function ψ . Suppose, for instance, we assume that both $\psi(\epsilon)$ and $\epsilon\psi'(\epsilon)$ are bounded. Then lemma 1 continues to hold if f is a function which can be expanded in a power series with radius of convergence exceeding 1. This includes harmonic polynomials as special cases, that is, functions of the form

$$f(x) = b_0 + \sum_{k=1}^p \{a_k \sin(2\pi kx) + b_k \cos(2\pi kx)\}.$$

The basis for the above claim is the theorem of Tompkins (1971) which implies that the law of the iterated logarithm (required in the proof of lemma 1) holds under these modified conditions.

(2) The independence of the ϵ_i was required only in Lemma 1 to justify application of certain iterated logarithm laws to the sequences $\psi(\epsilon_i)$ and

$\epsilon_i \psi(\epsilon_i)$. Now these laws can be shown to hold also for many dependent, stationary sequences and, consequently, Theorem 1 will hold for such sequences too. To be more specific, suppose the series $\{\epsilon_i\}$ is a stationary ϕ -mixing with zero mean and $E(|\epsilon_i|^{2+\delta}) < \infty$ for some $\delta > 0$. Suppose also that the mixing coefficients φ_n satisfy

$$\sum_{n \geq 1} \varphi_n^{\frac{1}{2}} < \infty$$

and that

$$\tau^2 \equiv \sum |l| < \infty \quad \text{cov}(\epsilon_i, \epsilon_{i+1})$$

is absolutely convergent and positive. For any function f such that $E[f(\epsilon_i)] = 0$ and $\text{var}[f(\epsilon_i)] < \infty$, it then holds that

$$(12) \quad \sum_{i=1}^n f(\epsilon_i) = \sum_{i=1}^n \eta_i + O(n^{\frac{1}{2}-\xi}) \quad \text{a.s.}$$

for some $\xi > 0$ and a sequence of independent normal $(0, \tau^2)$ random variables, see Phillip and Stout (1975). From (12) it follows that any iterated logarithm law which holds for the independent sequence $\{f(\frac{i}{n})\eta_i; 1 \leq i \leq n\}$, holds also for the dependent sequence $\{f(\frac{i}{n})\epsilon_i; 1 \leq i \leq n\}$. ■

ACKNOWLEDGEMENT

The authors wish to thank Dr. Tertius de Wet of the Institute for Maritime Technology for useful discussions on M -estimators.

BIBLIOGRAPHY

- BRILLINGER, D.R. (1975). *Time series: Data analysis and theory*. San Francisco. Holden-Day. Inc.
- CARROLL, R.J. (1978). On almost sure expansions for M -estimators. *Ann. Statist.*, 6, 14-318.

- CHOW, Y.S. & TEICHER, H. (1978). *Probability theory: Independence, interchangability, martingales*. Springer-Verlag, Heidelberg.
- FRESEN, J., LOMBARD, F., & DE WET, T. (1992). Residual processes in robust regression. In preparation.
- DUNFORD, N. & SCHWARTZ, J.T. (1963). *Linear operators*. Part I & II. New York. Interscience Publishers Inc.
- HUBER, P. (1981). *Robust Statistics*. New York: John Wiley and Sons.
- PHILLIP, W. & STOUT, W. (1975). Almost sure invariance principles for partial sums of weakly dependent random variables. *Mem. Amer. Math. Soc.*, No 161
- TOMKINS, R.J. (1971). An iterated logarithm theorem for some weighted averages of independent random variables. *Ann. Math. Statist.*, **42**, 760-763.